From: http://www.easyrgb.com/index.php?X=MATH

```
// RGB -> CMY
// RGB values from 0 to 255
// CMY results from 0 to 1

C = 1 - ( R / 255 )
M = 1 - ( G / 255 )
Y = 1 - ( B / 255 )

// CMY -> RGB
// CMY values from 0 to 1
// RGB results from 0 to 255

R = ( 1 - C ) * 255
G = ( 1 - M ) * 255
B = ( 1 - Y ) * 255

// CMY -> CMYK
// CMYK and CMY values from 0 to 1

var_K = 1

if ( C < var_K )   var_K = C
if ( M < var_K )   var_K = M
if ( Y < var_K )   var_K = Y
if ( var_K == 1 ) { //Black
   C = 0
   M = 0
   Y = 0
}
else {
   C = ( C - var_K ) / ( 1 - var_K )
   M = ( M - var_K ) / ( 1 - var_K )
   Y = ( Y - var_K ) / ( 1 - var_K )
}
K = var_K


// CMYK -> CMY
// CMYK and CMY values from 0 to 1

C = ( C * ( 1 - K ) + K )
M = ( M * ( 1 - K ) + K )
Y = ( Y * ( 1 - K ) + K )
```

```
// RGB -> HSV

var_R = ( R / 255 )                     //RGB from 0 to 255
var_G = ( G / 255 )
var_B = ( B / 255 )
var_Min = min( var_R, var_G, var_B )    //Min. value of RGB
var_Max = max( var_R, var_G, var_B )    //Max. value of RGB
del_Max = var_Max - var_Min             //Delta RGB value

V = var_Max

if ( del_Max == 0 )                     //This is a gray, no chroma...
{
   H = 0                                //HSV results from 0 to 1
   S = 0
}
else                                    //Chromatic data...
{
   S = del_Max / var_Max

   del_R = ( ( ( var_Max - var_R ) / 6 ) + ( del_Max / 2 ) ) / del_Max
   del_G = ( ( ( var_Max - var_G ) / 6 ) + ( del_Max / 2 ) ) / del_Max
   del_B = ( ( ( var_Max - var_B ) / 6 ) + ( del_Max / 2 ) ) / del_Max

   if      ( var_R == var_Max ) H = del_B - del_G
   else if ( var_G == var_Max ) H = ( 1 / 3 ) + del_R - del_B
   else if ( var_B == var_Max ) H = ( 2 / 3 ) + del_G - del_R

   if ( H < 0 ) ; H += 1
   if ( H > 1 ) ; H -= 1
}

// HSV -> RGB

if ( S == 0 )                       //HSV from 0 to 1
{
   R = V * 255
   G = V * 255
   B = V * 255
}
else
{
   var_h = H * 6
   if ( var_h == 6 ) var_h = 0      //H must be < 1
   var_i = int( var_h )             //Or ... var_i = floor( var_h )
   var_1 = V * ( 1 - S )
   var_2 = V * ( 1 - S * ( var_h - var_i ) )
   var_3 = V * ( 1 - S * ( 1 - ( var_h - var_i ) ) )

   if      ( var_i == 0 ) { var_r = V     ; var_g = var_3 ; var_b = var_1 }
   else if ( var_i == 1 ) { var_r = var_2 ; var_g = V     ; var_b = var_1 }
   else if ( var_i == 2 ) { var_r = var_1 ; var_g = V     ; var_b = var_3 }
   else if ( var_i == 3 ) { var_r = var_1 ; var_g = var_2 ; var_b = V     }
   else if ( var_i == 4 ) { var_r = var_3 ; var_g = var_1 ; var_b = V     }
   else                   { var_r = V     ; var_g = var_1 ; var_b = var_2 }

   R = var_r * 255                  //RGB results from 0 to 255
   G = var_g * 255
   B = var_b * 255
}
```